

Abstract Algebra II Homework 4

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Lemma 1.1. *Suppose G is a finite abelian group, $x, y \in G$, and $|x| = r$ and $|y| = s$. Then there exists an element of G which has order $\text{lcm}(r, s)$.*

Proof. First, suppose that $\gcd(r, s) = 1$, then $\text{lcm}(r, s) = \frac{rs}{\gcd(rs)} = rs$. Given that $|x| = r$ and $|y| = s$, consider $z = xy$, then $z^{rs} = x^{rs}y^{rs} = e$, we have $|z| \mid rs$ (so $|z| \leq rs$).

Let $|z| = m$, so $e = e^s = z^{ms} = x^{ms}y^{ms} = x^{ms}$, we have $r \mid ms$. Similarly, $e = e^r = z^{mr} = x^{mr}y^{mr} = y^{mr}$, we have $s \mid mr$. But $\gcd(r, s) = 1$, so $r \mid m$ and $s \mid m$, the result implies $rs \mid m$. Hence, $|m| \geq rs$. But also $|m| \leq rs$, so $|z| = m = rs = \text{lcm}(r, s)$.

We now consider the general case, we can decompose r as a product $r = r_1 r_2 r_3 r_4$ as follows:

- r_1 = the product of all prime factors of r which aren't prime factors of s .
- r_2 = the product of all prime factors of r which have equal power in r and s .
- r_3 = the product of all prime factors of r which have higher power than those in s .
- r_4 = the product of all prime factors of r which have lower power than those in s .

Define $s = s_1 s_2 s_3 s_4$ similarly, with r and s in exchanged places. And we have $\text{lcm}(r, s) = r_1 r_2 r_3 s_1 s_3$ and $\gcd(r_1 r_2 r_3, s_1 s_3) = 1$. Now $|x^{r_4}| = r_1 r_2 r_3$ and $|y^{s_2 s_4}| = s_1 s_3$, if we consider the two elements: x^{r_4} and $y^{s_2 s_4}$, $|x^{r_4} y^{s_2 s_4}| = r_1 r_2 r_3 s_1 s_3 = \text{lcm}(r, s)$ (trivially $x^{r_4} y^{s_2 s_4} \in G$) by the first part of the proof. So there exists an element of G which has order $\text{lcm}(r, s)$. $\square$

Let $|K^*| = m$. Suppose $\alpha \in K^*$ has the maximal order under multiplication over all elements in K^* . Let $|\alpha| = k$, by Lagrange's theorem, $k \mid m$ and $k \leq m$.

Consider another elements $\beta \in K^*$ with order r , by lemma 1.1, there exists an element of order $\text{lcm}(r, k) \geq k$, but due to the maximality, we must have $\text{lcm}(r, k) = k$. Clearly, $r \mid k$, and $\beta^k = e$ (where e is the multiplicative identity in K^*). Since β is chosen arbitrarily, we have every element's k -th power is the identity, i.e., satisfies the equation $x^k - 1 = 0$. Therefore, $x^k - 1 = 0$ has at least m distinct roots, and a polynomial of degree k over some fields has at most k roots in that field. So we have $m \leq k$, combined with our previous result we know $m = k$.

Since $|\alpha| = m = |K^*|$, α generates the whole group K^* (Here, the group is defined under multiplication, since in the multiplicative group of K , the zero is excluded, every element is unit, thus having its inverse. The closedness is trivial since the K is a field, where $K = K^* \cup \{0\}$), K^* is cyclic.

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Let E be n -dimensional extension of the finite field K (Let $|K| = q$). Because E acts like an n -dimensional vector space over K (which has q elements), the field E must contain exactly q^n elements.

The nonzero elements of E form a multiplicative group $E^\times = E \setminus \{0\}$, it contains $q^n - 1$ elements, i.e., $|E^\times| = q^n - 1$.

By Lagrange's Theorem, the order of every element in $E^\times$ must divides the $|E^\times|$. Hence, if $\alpha \in E^\times$, we have $|\alpha| \mid (q^n - 1)$, thus $\alpha^{q^n-1} = 1 \Rightarrow \alpha^{q^n-1} - 1 = 0 \Rightarrow \alpha^{q^n} - \alpha = 0$.

Hence, every element of $E^\times$ is the root of $x^{q^n} - x = 0$, even the zero satisfies it. We obtain that every element of E is a root of $x^{q^n} - x = 0$, also every polynomial of degree q^n has at most q^n distinct, which is equal to the order of E . Therefore, E is the splitting field of $x^{q^n} - x$.

Since both K_1 and K_2 are the splitting field of the same polynomial $(x^{q^n} - x)$, they are isomorphic.